Decision Tree

- DT
- Maths
  - Entropy
- Scratch
- SC / position of anything
- sklearn → DT

Feature

Age	Car	Job	Home	Class
young	yes	yes	own	yes
young	yes	yes	rent	yes
young	yes	no	own	yes
young	yes	no	rent	yes
young	no	yes	own	yes
young	no	yes	rent	yes
young	no	no	own	no
young	no	no	rent	no
old	yes	yes	own	yes
old	yes	yes	rent	yes
old	yes	no	own	yes
old	yes	no	rent	yes
old	no	yes	own	yes
old	no	yes	rent	yes
old	no	no	own	no
old	no	no	rent	no

Class → dog

Depends upon Age

age = young

If age = young
  If Car = yes: dog
  else: cat
else: if Job = yes: dog
  else: if Home = own: dog
  else: cat

In decision Tree → Nodes/Conditions

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200 data point

y=	100
y=	100

→ completely impure.

y=	100
y=	50
y=	50

→ impure

y=	100
y=	90
y=	10

→ slightly impure

y=	100
y=	100

→ pure node

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age > 40
  yes → only 1 class
  no → class is "mixed"

y=	100
y=	50
y=	50

age > 40
  yes → class 100 100
  no → class 100 100